Fiscal Year:	FY 2025	**Task Last Updated:**	FY 02/17/2025
PI Name:	Fernandez-Pello, Carlos Ph.D.		
Project Title:	Material Ignition and Suppression Test (MIST) in Space Exploration Atmospheres		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94720-1740	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Physical Sciences Unsolicited
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No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 09/30/2026 per the PI (Ed., 9/16/25). NOTE: Start dates and end dates were updated per J. McQuillen/NASA-GRC. Original dates were 02/09/2022 - 02/28/2025 (Ed., 12/23/24).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Gollner, Michael (University of California, Berkeley)		
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Task Description:	NOTE: Continuation of NNX10AE01G, Material Ignition and Suppression in Space Exploration Atmospheres, with the same principal investigator, Prof. Carlos Fernandez-Pello. Central Objectives: A study is underway to conduct an experimental and theoretical research program on the effect of an external radiant flux on the ignition, subsequent flame spread, and fire suppression of a representative combustible material in space exploration atmospheres (SEA). The near-term objective of the research is to develop a ground based Material Ignition and Suppression Test (MIST) apparatus that reflects the environments that are expected in space-based facilities and is similar to the Solid Fuel Ignition and Extinction / Material Ignition and Suppression Test (SoFIE/MIST) apparatus that will be installed in the International Space Station (ISS). The ground based MIST apparatus has already been built and tested during the research program for NASA Grant #NNX10AE01G, and the apparatus is currently being used to obtain a flammability boundary, ignition, flame spread and extinction, of a polymeric material, specifically PMMA, which can be viewed as a baseline material for fire testing. As the microgravity data from the SoFIE/MIST experiments is available, it will be analyzed and compared with the normal gravity data to determine the effect of reduced gravity on the flammability of materials in SEA. Methods/techniques: The experimental apparatus consists of a small-scale rectangular flow duct that has a cylindrical fuel sample at the axis of the duct and has four radiant heaters placed in the corners of the duct that radiate the sample surface with a predetermined radiant flux. An electrical coil ignitor is placed at the trailing edge of the sample to ignite the sample and initiate flame spread. For SEA testing in normal gravity, the apparatus is placed inside a pressure chamber that resembles that of the Combustion Integrated Rack (CIR) in the ISS. An oxidizer gas, at SEA oxygen concentrations and pressures, flows at low velocity through the length of the duct. The fuel samples are rods of black PMMA of different diameters. The radiant heaters provide a radiant flux over the sample surface that increases its temperature and enhances the rate of flame spread. The ignitor provides the necessary energy to ignite the polymeric samples. The tests provide information about the effect of a radiant flux on the ignition, flame spread, and burning rate at a given oxygen concentration and ambient pressure. This will be followed by the reduction of the O2 concentration with N2 or CO2 until extinction occurs. The result will provide information about the suppression characteristics of inert gases in SEA and of the effect of SEA on the Limiting Oxygen Index (LOI) of the material when exposed to an external radiant flux. The experimental task will be complemented with a theoretical task that includes both analytical and numerical modeling. The analytical modeling uses simplified formulas of flame spread previously developed that incorporate the effect of the external radiant flux on the flame spread rate through the energy received by the fuel rod at a particular time. The numerical model will use a solid combustible pyrolysis numerical code developed in the Principal Investigator (PI) laboratory, together with a computational fluid dynamics (CFD) code simulating the reacting gas phase. The microgravity experiments conducted under the SoFIE project will be used to validate the theoretical model. It is the objective that the theoretical model will prevent the need to conduct extensive experimental testing of materials in microgravity. Perceived Significance: There is insufficient knowledge about fire behavior of materials in environments expected in space-based facilities, i.e., SEA (micro/partial-gravity, low velocity flow, reduced pressure, and elevated oxygen concentrations). Nor is there a testing methodology specifically developed to determine the fire hazards of materials under those conditions. It is the long-term objective of this project to provide additional information about the effect of SEA on the flammability (ignition, flame spread, and heat release rate) and suppression of materials, and guidance to interpret and extend current NASA testing procedures to environments expected in space-based facilities.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The project is intended to enhance the ability to predict ignition, burning (flame spread and mass burning), and extinction characteristics of materials in their end use configuration. This approach is called performance based design in terrestrial fire protection engineering. Computational support for this approach requires detailed information about material thermal and combustion properties. Our work on MIST will use experimental data to validate computer models attempting to predict fire behavior using material property estimation techniques. These techniques are currently a focus of research in fire protection engineering.
	Emphasis is given to the differences in the limiting oxygen volume fraction for extinction of flame spread and fuel regression in an enclosure at sub-atmospheric pressures, under normal and microgravity. More detailed information about the research conducted during this reporting period can be found in the cited publications. 1. Experiments on the opposed flame spread and fuel regression over PMMA rods in normal and microgravity The primary motivation of the work is to provide information about the extinction of solid fuels in atmospheres expected in future spacecraft. Experiments with cylindrical black poly(methyl methacrylate) (PMMA) samples are conducted at sub-atmospheric pressures in a sealed combustion chamber on the Combustion Integrated Rack (CIR) of the International Space Station (ISS), and in a similar chamber in normal gravity. The oxygen concentration in the chamber is left to decrease naturally at constant pressure as the PMMA burns until extinction. The path to extinction of the opposed spread of flames and the fuel regression behind the flame is followed and their limiting oxygen volume fraction at extinction (LOC) recorded. It is found that there is an LOC at which the flame stops spreading, referred to here as the limiting oxygen volume fraction for flame spread (LOC-FS), and another at lower oxygen concentration at which the regressing flame extinguishes (LOC-FR). It is also found that both LOCs decrease with ambient pressure and are significantly lower in microgravity than at Earth. The findings of this work show that current Earth-based flammability standards are not conservative enough and are insufficient for environments that future spacecraft are expected to operate in. Flame Spread Behavior: After the ignition of the PMMA rod is established, a flame starts spreading along the solid fuel surface against the opposed forced gas flow. In normal gravity, the opposed flow is a mixed forced and buoyant flow, with the velocity of the latter depending on the ambient pressure and oxygen concentration. In microgravity (μg), the gas flow is forced. As the flame spreads along the rod's surface, it leaves behind a regressing burning surface of the shape of a cone. During steady flame spread, angle of the cone is determined by the ratio of the flame spread rate and the flame regression rate. When the flame stops spreading but the PMMA continues burning (regressing), the cone angle is determined by the fuel regression rate. Since the chamber is closed, as the flame spreads over the sample, the chamber's oxygen concentration decreases due to

Task Progress:	the combustion of the PMMA, which causes the flame spread (FS) rate, and the fuel regression rate, to also decrease. The flame characteristics also change as the chamber's oxygen concentration is depleted. After ignition, the flame begins to spread along the PMMA rod surface against the gas flow. Subsequently, the flame advances over the surface of the PMMA at a rate that decreases as burning progresses. During this phase, the solid material is mostly consumed radially, causing the conical shape of the PMMA to become steeper. At a certain point, the flame ceases to propagate; this transition limit is the LOC for Flame Spread, (LOC-FS). Since the PMMA downstream continues to burn, the oxygen in the chamber continues to decrease, and the flame retracts somewhat until it anchors itself in the stagnation region behind the forward pyrolysis front. The PMMA continues to be consumed radially, and the height of the formed cone decreases significantly compared to the FS phase. As the oxygen concentration decreases, the flame in the fuel regression zone decreases and eventually extinguishes. This is the LOC for fuel regression, LOC-FR. The results of the tests show that, both in 1-g and µg, as the oxygen concentration gradually decreases due to the combustion of the PMMA, the flame spread rate also decreases until it reaches an oxygen volume fraction where the flame no longer spreads. As indicated above, this oxygen threshold is defined here as the Limiting Oxygen Concentration (Volume Fraction) for Flame Spread (LOC-FS). It is observed that the flame spreads significantly slower in microgravity than in normal gravity, taking three times longer to reach the LOC-FS. The oxygen is also consumed slower in microgravity than in normal gravity. For the rods of 12.7 mm diameter, it is found that the LOC-FS is approximately 18.8% in normal gravity, while it is ~16.2% in microgravity at a chamber pressure of 100 kPa. It is also observed that after the LOC-FS is reached, as the oxygen is consumed further, there is a transition period where the flame leading edge moves gradually backward, in what appears to be a weak blow-off extinction. In normal gravity, the flame eventually stops receding and attaches to the cone behind the pyrolysis front. As the PMMA surface keeps regressing, the cone angle becomes steeper, the oxygen concentration becomes smaller, and the flame weaker, until it finally extinguishes. This oxygen threshold is defined here as the Limiting Oxygen Concentration (Volume Fraction) for Fuel Regression (LOC-FR). The LOC-FR threshold is the oxygen concentration normally recorded in the LOI apparatus since it reports the visual extinction of the flame. It is important to notice that the LOC-FR is significantly lower than the LOC-FS. This is relevant since a spreading flame has an influence on the growth of fire while the burning regression of the fuel influences the heat release rate. From 1-g and µg experiments at pressures of 80, 60, and 40 kPa, it is observed that reducing the chamber ambient pressure affects the flame's behavior across all measured parameters. Specifically, the LOC for the flame spread, and regression thresholds, are notably lower at reduced pressures. These are important results because they show that PMMA is more flammable in microgravity than in normal gravity, at least for the current experimental conditions. 2. Modeling Extinction of Opposed Flame Spread over PMMA Rods During this reporting period, a simplified analytical model of the oxygen volume fraction (LOC), leading to the extinction of opposed flame spread over combustible materials, was developed. The analysis is based on the condition that caused the flame spread rate, or the fuel regression rate, to become zero, i.e., no burning. The analysis predicts two oxygen volume fraction boundaries for extinction: a "heat transfer" boundary and a "chemical kinetics" boundary. The former is given by the conditions needed to heat the solid to its pyrolysis temperature, and the latter by the conditions needed for the gas phase reaction to be maintained. The combination of the two boundaries provides an overall boundary for the oxygen volume fraction for extinction as a function of material properties, gas flow velocity, ambient pressure, gravity level, and external heating. The analysis shows that at high flow velocity and/ or elevated pressures flame extinction (LOC) occurs due to chemical kinetic effects. This limit is often referred to as "flame "blow-off.". However, it is also possible to appreciate that at very low flow velocities like those experienced in reduced gravity or microgravity, where the buoyant flow would be small or negligible, the blow-off limit could occur at low flow velocities, particularly at low ambient pressure. Also, at very low flow velocities in reduced gravity, the heat flux from the flame to the solid would be small; because the flame standoff distance would be large, the surface radiative heat losses can become critical in determining the LOC. Under these conditions, the transport of the ambient oxygen to the flame would be reduced, which would decrease the flame temperature, and in turn cause its extinction (LOC). This limit has been also referred to as the diffusion limit.
Bibliography Type:	Description: (Last Updated: 02/15/2026)
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Articles in Peer-reviewed Journals	Liveretou C, Scudiere C, Rivera J, Etzenbach L, Thomsen M, Fernandez-Pello C, Gollner MJ, Olson S, Ferkul P. "The limiting oxygen volume fraction for opposed flame spread extinction." Proceedings of the Combustion Institute. 2024 Jan 1;40(1-4):105224. https://doi.org/10.1016/j.proci.2024.105224 , Jan-2024
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